

# **CO1-AT2 DATA WRANGLING PRACTICAL ASSIGNMENT**

Practical Record

**Course Code:** DSA0502

**Course Title:** QUERY PROCESSING FOR DATA SCIENCE

**Title:** Data Wrangling Practical Assignment

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## **Experiment 30: Employee Information from XML**

### **Aim**

To read employee information from an XML file, display employee names and departments, count employees in each department, and export the results into a CSV file using Python.

### **Algorithm**

1. Start the program.
2. Import the required libraries (xml.etree.ElementTree and pandas).
3. Load and parse the XML file (employees.xml).
4. Extract employee names and departments.
5. Display the employee names and departments.
6. Count the number of employees in each department.
7. Store the department counts in a DataFrame.
8. Export the DataFrame to a CSV file.
9. Display a success message.

## PYTHON CODE

```
task30_employee_xml (1).py X employees.xml :: ▶ ↺ ⬇ ⬆ ↻ □
task30_employee_xml (1).py > read_employees
1  """
2  Task 30: Employee Information from XML
3  -----
4  Reads employee records from an XML file (employees.xml), displays each
5  employee's name and department, counts how many employees belong to each
6  department, and exports a summary report to a CSV file.
7
8  Input : employees.xml
9  Output: employee_department_summary.csv
10 """
11
12 import xml.etree.ElementTree as ET
13 import csv
14 from collections import Counter
15
16
17 def read_employees(xml_path):
18     """Parse the XML file and return a list of (name, department) tuples."""
19     tree = ET.parse(xml_path) # Load and parse the XML file into a tree xml_path = 'employees.xml'
20     root = tree.getroot() # Get the root element (<Employees>)
21
22     employees = []
23     for emp in root.findall("Employee"): # Loop through every <Employee> node
24         name = emp.find("Name").text.strip()
25         department = emp.find("Department").text.strip()
26         employees.append((name, department))
27     return employees
28
29
30 def display_employees(employees):
31     """Print each employee's name and department to the console."""
32     print("Employee Name      Department")
33     print("-" * 35)
34     for name, department in employees:
35         print(f"{name:<20} {department}")
36
37
```

## Output

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
b\task30_employee_xml (1).py"
Name Department
0 John HR
1 Alice IT
2 Bob IT
3 David Finance
4 Emma HR

Employee Count by Department:
Department
HR 2
IT 2
Finance 1
Name: count, dtype: int64

CSV file created successfully.
PS C:\Users\vigne\OneDrive\Documents\co1 at2 b> |
```

## Experiment 31: Hospital Patient Record Cleanup

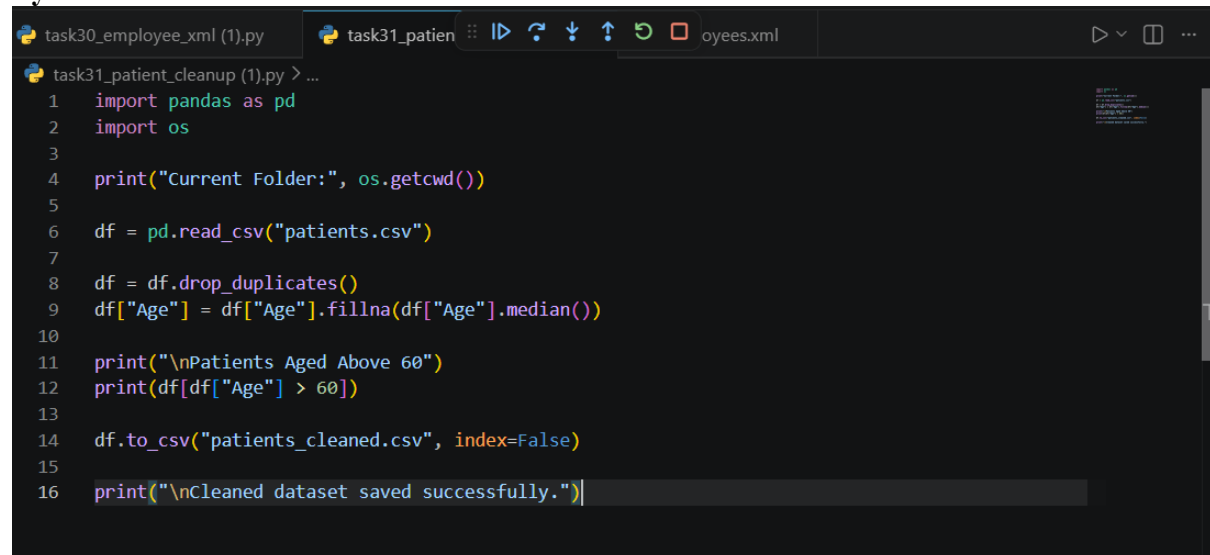
### Aim

To clean hospital patient records by removing duplicate entries, replacing missing age values with the median age, displaying patients aged above 60 years, and saving the cleaned dataset using Python.

### Algorithm

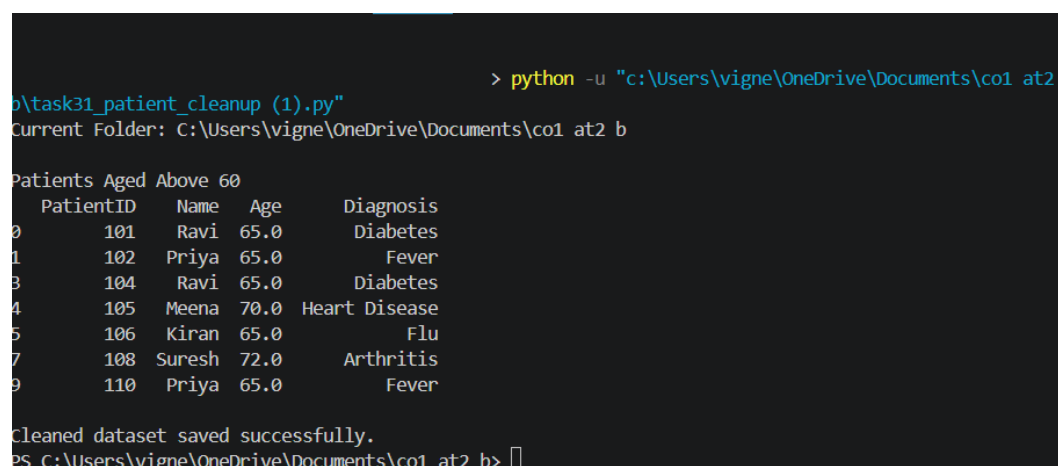
1. Start the program.
2. Import the Pandas library.
3. Read the patients.csv file into a DataFrame.
4. Remove duplicate patient records using `drop_duplicates()`.
5. Calculate the median value of the Age column.
6. Replace missing age values with the median age using `fillna()`.
7. Display the records of patients whose age is greater than 60.
8. Save the cleaned dataset as `patients_cleaned.csv`.
9. Display a success message.
10. Stop the program.

### Python code



```
task31_patient_cleanup (1).py > ...
1  import pandas as pd
2  import os
3
4  print("Current Folder:", os.getcwd())
5
6  df = pd.read_csv("patients.csv")
7
8  df = df.drop_duplicates()
9  df["Age"] = df["Age"].fillna(df["Age"].median())
10
11 print("\nPatients Aged Above 60")
12 print(df[df["Age"] > 60])
13
14 df.to_csv("patients_cleaned.csv", index=False)
15
16 print("\nCleaned dataset saved successfully.")
```

### Output



```
> python -u "c:\Users\vigne\OneDrive\Documents\co1 at2 b\task31_patient_cleanup (1).py"
Current Folder: C:\Users\vigne\OneDrive\Documents\co1 at2 b

Patients Aged Above 60
  PatientID  Name  Age  Diagnosis
0        101   Ravi  65.0    Diabetes
1        102  Priya  65.0      Fever
3        104   Ravi  65.0    Diabetes
4        105  Meena  70.0  Heart Disease
5        106  Kiran  65.0      Flu
7        108  Suresh  72.0   Arthritis
9        110  Priya  65.0      Fever

Cleaned dataset saved successfully.
PS C:\Users\vigne\OneDrive\Documents\co1 at2 b>
```